EXPERIMENT	INSTALL AND RUN HIVE, THEN USE HIVE TO MANIPULATE THE DATABASE	ROLL NO	23DI16	
		DAY	MONTH	YEAR
8				

AIM:

To Install and Run Hive, then use Hive to manipulate the database.

COMPONENTS REQUIRED:

S.No	Component	Quantity
1	Running PC	1
2	Ubuntu OS	1

PROCEDURE:

Step 1: Switch to Hadoop user Su

– hdoop

Step 2: Use Apache archive url

wget https://archive.apache.org/dist/hive/hive-3.1.2/apache-hive-3.1.2-bin.tar.gz

```
hdoop@ubuntumi:~$ wget https://archive.apache.org/dist/hive/hive-3.1.2/apache-h
ive-3.1.2-bin.tar.gz
--2026-01-20 20:11:28--  https://archive.apache.org/dist/hive/hive-3.1.2/apache
-hive-3.1.2-bin.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1
a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... fa
iled: Connection refused.
Connecting to archive.apache.org (archive.apache.org)|2a01:4f9:1a:a084::2|:443.
.. connected.
HTTP request sent, awaiting response... 200 OK
Length: 278813748 (266M) [application/x-gzip]
Saving to: 'apache-hive-3.1.2-bin.tar.gz'

apache-hive-3.1.2-b 100%[===================>] 265.90M  9.82MB/s    in 34s

2026-01-20 20:12:24 (7.86 MB/s) - 'apache-hive-3.1.2-bin.tar.gz' saved [2788137
48/278813748]
```

Step 3: Extract Hive tar -xvzf

apache-hive-3.1.2-bin.tar.gz **Step 4:**

Edit .bashrc nano ~/.bashrc

Add only these lines at the end

export HIVE_HOME=/home/hdoop/apache-hive-3.1.2-bin export PATH=$PATH:$HIVE_HOME/bin

```
export PIG_CLASSPATH=$HADOOP_HOME/etc/Hadoop
export HIVE_HOME=/home/hdoop/apache-hive-3.1.2-bin
export PATH=$PATH:$HIVE_HOME/bin
```

Step 5: Reload .bashrc source

~/.bashrc

Step 6: Create Required HDFS Directories

make sure Hadoop is running hdfs dfs -

mkdir /tmp hdfs dfs -chmod g+w /tmp

hdfs dfs -mkdir -p /user/hive/warehouse hdfs

dfs -chmod g+w /user/hive/warehouse

```
hdoop@ubuntumi:~$ hdfs dfs -chmod g+w /user/hive/warehouse
hdoop@ubuntumi:~$ hdfs dfs -ls /
Found 3 items
drwxr-xr-x   - hdoop supergroup          0 2026-01-20 19:44 /pig_data
drwxrwxr-x   - hdoop supergroup          0 2026-01-20 20:04 /tmp
drwxr-xr-x   - hdoop supergroup          0 2026-01-20 20:15 /user
```

Step 7: Start Hive

Hive

```
hdoop@dit-OptiPlex-3060:~$ hive
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/hdoop/apache-hive-3.1.2-bin/lib/log4j-slf4j-impl-2.10.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/hdoop/hadoop-3.4.0/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Hive Session ID = d0504aa5-b23f-417c-8692-b1344a546fc2

Logging initialized using configuration in jar:file:/home/hdoop/apache-hive-3.1.2-bin/lib/hive-common-3.1.2.jar!/hive-log4j2.properties Async: true
Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. spark, tez) or using Hive 1.X rele
ases.
hive> Aagash , dwarakesh , shyam
```

RESULT:

Thus, to Install and Run Hive, then use Hive to manipulate the database has executed successfully.

EXPERIMENT	INSTALL, DEPLOY & CONFIGURE APACHE SPARK CLUSTER. RUN APACHE SPARK APPLICATIONS USING SCALA.	ROLL NO	23DI16	
		DAY	MONTH	YEAR
9				

AIM:

To Install,Deploy & configure Apache Spark Cluster. Run apache spark applications using Scala.

COMPONENTS REQUIRED:

S.No	Component	Quantity
1	Running PC	1
2	Ubuntu OS	1

PROCEDURE:

Step 1: Download apache spark wget https://archive.apache.org/dist/spark/spark-3.5.3/spark-3.5.3-bin-hadoop3-scala2.13.tgz

```
hdoop@ubuntumi:~$ wget https://archive.apache.org/dist/spark/spark-3.5.3/spark-
3.5.3-bin-hadoop3-scala2.13.tgz
--2026-01-20 20:30:29--  https://archive.apache.org/dist/spark/spark-3.5.3/spar
k-3.5.3-bin-hadoop3-scala2.13.tgz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1
a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... fa
iled: Connection refused.
Connecting to archive.apache.org (archive.apache.org)|2a01:4f9:1a:a084::2|:443.
.. connected.
HTTP request sent, awaiting response... 200 OK
Length: 409868122 (391M) [application/x-gzip]
Saving to: 'spark-3.5.3-bin-hadoop3-scala2.13.tgz'

spark-3.5.3-bin-had 100%[===================>] 390.88M  6.94MB/s    in 3m 45s

2026-01-20 20:34:37 (1.74 MB/s) - 'spark-3.5.3-bin-hadoop3-scala2.13.tgz' saved
 [409868122/409868122]
```

Step 2: Move it to the system location mv spark-3.5.3-bin-hadoop3-scala2.13 ~/spark

Set environment variable Open

your bash profile:

nano ~/.bashrc

Add these lines at the end:
export SPARK_HOME=/opt/spark export

PATH=$PATH:$SPARK_HOME/bin

Step 3: Apply the changes source

~/.bashrc

Step 4: Now you can run Spark

Spark shell

```
hdoop@dit-OptiPlex-3060:~$ nano ~/.bashrc
hdoop@dit-OptiPlex-3060:~$ spark-shell
26/01/21 17:19:32 WARN Utils: Your hostname, dit-OptiPlex-3060 resolves to a loopback address: 127.0.1.1; using 10.1.130.62 instead (on interface enp2s0)
26/01/21 17:19:32 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 3.5.3
      /_/

Using Scala version 2.13.8 (OpenJDK 64-Bit Server VM, Java 11.0.29)
Type in expressions to have them evaluated.
Type :help for more information.
26/01/21 17:19:35 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Spark context Web UI available at http://10.1.130.62:4040
Spark context available as 'sc' (master = local[*], app id = local-1768996175754).
Spark session available as 'spark'.

scala> Aagash , dwarakesh , shyam
```

RESULT:

Thus, to Install, Deploy & configure Apache Spark Cluster. Run apace spark applications using Scala has executed successfully.

EXPERIMENT	DATA ANALYTICS USING APACHE SPARK IN ANY DATASET	ROLL NO	23DI16	
		DAY	MONTH	YEAR
10				

AIM:

To Data analytics using Apache Spark in any dataset.

COMPONENTS REQUIRED:

S.No	Component	Quantity
1	Running PC	1
2	Ubuntu OS	1

PROCEDURE:

Step 1: Start PySpark pyspark

```
hdoop@ubuntumi:~$ pyspark
Python 3.10.12 (main, Nov  4 2025, 08:48:33) [GCC 11.4.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
26/01/20 20:46:04 WARN Utils: Your hostname, ubuntumi resolves to a loopback ad
dress: 127.0.1.1; using 10.0.2.15 instead (on interface enp0s3)
26/01/20 20:46:04 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another
 address
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLev
el(newLevel).
26/01/20 20:46:07 WARN NativeCodeLoader: Unable to load native-hadoop library f
or your platform... using builtin-java classes where applicable
26/01/20 20:46:09 WARN Utils: Service 'SparkUI' could not bind on port 4040. At
tempting port 4041.
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 3.5.3
      /_/

Using Python version 3.10.12 (main, Nov  4 2025 08:48:33)
Spark context Web UI available at http://10.0.2.15:4041
Spark context available as 'sc' (master = local[*], app id = local-176892216975
```

Step 2: Check Spark version, python version, Spark master node,

sc.version sc.pythonVer sc.master

Step 3: Create a sample dataset in Python my_list = range(1, 10) # Creates a list of numbers from 1 to 9 Perform a simple transformation

squared_list_lambda = list(map(lambda x: x**2, my_list))

Print the result: print("The squared numbers are", squared_list_lambda)

```
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 3.5.3
      /_/

Using Python version 3.10.12 (main, Jan  8 2026 06:52:19)
Spark context Web UI available at http://10.1.130.62:4041
Spark context available as 'sc' (master = local[*], app id = local-1768996333662).
SparkSession available as 'spark'.
>>> sc.version
'3.5.3'
>>> sc.pythonVer
'3.10'
>>> sc.master
'local[*]'
>>> my_list = range(1, 10) # Creates a list of numbers from 1 to 9
>>> squared_list_lambda = list(map(lambda x: x**2, my_list))
>>> print("The squared numbers are", squared_list_lambda)
The squared numbers are [1, 4, 9, 16, 25, 36, 49, 64, 81]
>>> Aagash , dwarakesh , shyam
```

RESULT:

Thus, to Data analytics using Apache Spark in any dataset has executed successfully.

```
hdoop@ubuntumi:~$ wget https://archive.apache.org/dist/hive/hive-3.1.2/apache-h
ive-3.1.2-bin.tar.gz
--2026-01-20 20:11:28--  https://archive.apache.org/dist/hive/hive-3.1.2/apache
-hive-3.1.2-bin.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1
a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... fa
iled: Connection refused.
Connecting to archive.apache.org (archive.apache.org)|2a01:4f9:1a:a084::2|:443.
.. connected.
HTTP request sent, awaiting response... 200 OK
Length: 278813748 (266M) [application/x-gzip]
Saving to: 'apache-hive-3.1.2-bin.tar.gz'

apache-hive-3.1.2-b 100%[===================>] 265.90M  9.82MB/s    in 34s

2026-01-20 20:12:24 (7.86 MB/s) - 'apache-hive-3.1.2-bin.tar.gz' saved [2788137
48/278813748]
```

```
export PIG_CLASSPATH=$HADOOP_HOME/etc/Hadoop
export HIVE_HOME=/home/hdoop/apache-hive-3.1.2-bin
export PATH=$PATH:$HIVE_HOME/bin
```

```
hdoop@ubuntumi:~$ hdfs dfs -chmod g+w /user/hive/warehouse
hdoop@ubuntumi:~$ hdfs dfs -ls /
Found 3 items
drwxr-xr-x   - hdoop supergroup          0 2026-01-20 19:44 /pig_data
drwxrwxr-x   - hdoop supergroup          0 2026-01-20 20:04 /tmp
drwxr-xr-x   - hdoop supergroup          0 2026-01-20 20:15 /user
```

```
hdoop@dit-OptiPlex-3060:~$ hive
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/home/hdoop/apache-hive-3.1.2-bin/lib/log4j-slf4j-impl-2.10.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/hdoop/hadoop-3.4.0/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Hive Session ID = d0504aa5-b23f-417c-8692-b1344a546fc2

Logging initialized using configuration in jar:file:/home/hdoop/apache-hive-3.1.2-bin/lib/hive-common-3.1.2.jar!/hive-log4j2.properties Async: true
Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. spark, tez) or using Hive 1.X rele
ases.
hive> Aagash , dwarakesh , shyam
```

9a

```
hdoop@ubuntumi:~$ wget https://archive.apache.org/dist/spark/spark-3.5.3/spark-
3.5.3-bin-hadoop3-scala2.13.tgz
--2026-01-20 20:30:29--  https://archive.apache.org/dist/spark/spark-3.5.3/spar
k-3.5.3-bin-hadoop3-scala2.13.tgz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1
a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... fa
iled: Connection refused.
Connecting to archive.apache.org (archive.apache.org)|2a01:4f9:1a:a084::2|:443.
.. connected.
HTTP request sent, awaiting response... 200 OK
Length: 409868122 (391M) [application/x-gzip]
Saving to: 'spark-3.5.3-bin-hadoop3-scala2.13.tgz'

spark-3.5.3-bin-had 100%[===================>] 390.88M  6.94MB/s    in 3m 45s

2026-01-20 20:34:37 (1.74 MB/s) - 'spark-3.5.3-bin-hadoop3-scala2.13.tgz' saved
 [409868122/409868122]
```

```
hdoop@dit-OptiPlex-3060:~$ nano ~/.bashrc
hdoop@dit-OptiPlex-3060:~$ spark-shell
26/01/21 17:19:32 WARN Utils: Your hostname, dit-OptiPlex-3060 resolves to a loopback address: 127.0.1.1; using 10.1.130.62 instead (on interface enp2s0)
26/01/21 17:19:32 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
Welcome to
      ___          __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 3.5.3
      /_/

Using Scala version 2.13.8 (OpenJDK 64-Bit Server VM, Java 11.0.29)
Type in expressions to have them evaluated.
Type :help for more information.
26/01/21 17:19:35 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Spark context Web UI available at http://10.1.130.62:4040
Spark context available as 'sc' (master = local[*], app id = local-1768996175754).
Spark session available as 'spark'.

scala> Aagash , dwarakesh , shyam
```

10a

```
hdoop@ubuntumi:~$ pyspark
Python 3.10.12 (main, Nov  4 2025, 08:48:33) [GCC 11.4.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
26/01/20 20:46:04 WARN Utils: Your hostname, ubuntumi resolves to a loopback ad
dress: 127.0.1.1; using 10.0.2.15 instead (on interface enp0s3)
26/01/20 20:46:04 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another
 address
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLev
el(newLevel).
26/01/20 20:46:07 WARN NativeCodeLoader: Unable to load native-hadoop library f
or your platform... using builtin-java classes where applicable
26/01/20 20:46:09 WARN Utils: Service 'SparkUI' could not bind on port 4040. At
tempting port 4041.
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 3.5.3
      /_/

Using Python version 3.10.12 (main, Nov  4 2025 08:48:33)
Spark context Web UI available at http://10.0.2.15:4041
Spark context available as 'sc' (master = local[*], app id = local-176892216975
```

```
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 3.5.3
      /_/

Using Python version 3.10.12 (main, Jan  8 2026 06:52:19)
Spark context Web UI available at http://10.1.130.62:4041
Spark context available as 'sc' (master = local[*], app id = local-1768996333662).
SparkSession available as 'spark'.
>>> sc.version
'3.5.3'
>>> sc.pythonVer
'3.10'
>>> sc.master
'local[*]'
>>> my_list = range(1, 10) # Creates a list of numbers from 1 to 9
>>> squared_list_lambda = list(map(lambda x: x**2, my_list))
>>> print("The squared numbers are", squared_list_lambda)
The squared numbers are [1, 4, 9, 16, 25, 36, 49, 64, 81]
>>> Aagash , dwarakesh , shyam
```